

Social Intuition vs. Machine Reasoning: Anticipating Human-Robot Interaction from multiple modalities

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Abstract—Anticipating whether a person will interact from one’s own perspective is a highly intuitive task for humans, that relies on a combination of cues.

We investigate how humans perform at predicting a person’s intention to interact from a service robot’s point of view, using pose-only or full video input, then benchmark different lightweight pose-based models and state-of-the-art vision-language models.

We conducted our benchmark on the *HUI360* dataset on a fixed pilot subset of 100 test tracks (25 positive, 75 negative). We found that with pose-only input, human annotators outperform lightweight trained pose models but not by large margins (+0.08 in F1-Score). But when given full egocentric video with a target bounding box, human annotators perform substantially better and largely outperform the Vision-Language Models (+0.2 in F1-Score). We also compared VLMs of different size and under different input conditions, and found that the best results do not correlate with model size. Our result confirms that predicting interactions is a challenging task for social robots and that reasoning-capable models are necessary but their actual reasoning capabilities alone do not suffice to match the social intuition of humans.

Index Terms—human-robot interaction, anticipation, egocentric vision, vision-language models, nonverbal cues

I. INTRODUCTION

Social robots operating among people need reliable predictors of interaction and intention with enough anticipation to adequately plan motion and behaviors. In particular the *binary* prediction of whether someone will or will not interact with the robot is crucial for robots to address only those who are willing to interact.

From a computer-vision perspective, recent progress in video understanding using vision-language models makes them natural candidates for this task, but they remain computationally expensive to run and suffer from high latency, whereas on the other hand the very lightweight 2D-pose modality can be processed in real-time on edge devices.

We investigate how the interactant representation choice affects the prediction performance on the anticipation task, both for humans and vision models, and for the latter how different classes of machine learning approaches perform, under different modalities.

We build on the **HUI360** dataset and pose-learning baselines [1] with a fixed pilot benchmark: 100 test tracks (25 interaction-positive, 75 negative) and a one-second cut-off before the predicted interaction window. Human annotators label each track under modalities: **Pose** (2D-pose overlay on a static scene context) and $\mathbf{V}_{\text{bbox}}$ (egocentric video with a target bounding box).

Baselines trained with the **Pose** modality include reference sequence models trained end-to-end on sequences: an LSTM baseline, ST-GCN [2], and SkateFormer [3].

On the VLMs side we compare six models spanning on-device and API inference: Qwen3.5-0.8B and InternVL3-1B (served locally), Qwen3.5-35B-A3B and Qwen3.8-Max, and Gemini 3.5-Flash-Lite and Gemini 3.6-Flash, evaluated under three input conditions: $\mathbf{I}_c$ (image crop), $\mathbf{V} + \mathbf{I}_c$ (video with image reference), and $\mathbf{V}_{\text{bbox}}$ (video with per-frame bounding box).

We further ablate thinking mode, sampling rate, prompt length, and equirectangular vs. pinhole-camera style projection on two representative models.

II. RELATED WORK

Interaction anticipation and datasets. HUI360 [1] is the largest in-the-wild dataset for egocentric human-robot interaction anticipation; Earlier intent-prediction benchmarks are smaller or less reproducible. The People Approaching Robots Database (PAR-D) [4] aggregates data from two robots in public environments, including *Shutter*, a stationary robot photographer tracking human subjects with Azure Kinect sensors and from the ATC dataset with a mobile service robot in a shopping mall [5]. Human performance was assessed on PAR-D from short 3D skeleton visualizations (0.2 to 4 seconds) shown to crowd annotators (balanced samples). The results of [4] establish average human F1-scores between 0.65 and 0.86 with a noticeable variance of human accuracy depending on the observation length and the deployment context in each split.

VLMs for social and anticipatory perception. Recent HRI studies use video VLMs as commonsense proxies for when to engage, often triggered by lightweight preamble detectors [6].

In social navigation, VLM-Social-Nav [7] scores candidate robot actions with a vision-language model to produce socially compliant trajectories at inference time, without task-specific training on navigation datasets. At the same time some works on affect perception with VLMs have shown that models still struggle with emotion recognition from videos, suffering from inherent and prompt-induced biases [8], [9] as well as the lack of high frequency temporal modelling [9]. In [10] VLMs to human anticipatory judgment are compared on outcome prediction from contextual video and from observer’s facial reactions, finding strong context-based performance but weaker use of social cues.

III. METHODS

A. Dataset

HUI360 [1] is an egocentric 360° dataset from Shelfy, a static tray-holding robot in nine indoor environments. Persons are detected and tracked with SAM2 [11] on equirectangular frames; interactions are auto-annotated from masks and the tray zone, then manually reviewed. The HUI split has 4310 samples (375 interactions), we do not use the complementary *SSUP-A* split (>28,000 samples; a different robot and domain). Body keypoints come from ViTPose [12] (17) and Sapiens [13] (308); **Pose** models are retrained on HUI-Train with 17 COCO joints.

A *track* is one followed person over a short window in the 360° view. We evaluate 100 HUI-Test tracks (two environments unseen by the **Pose** models): 25 *positive* (the person will physically pick or place an item on the tray) and 75 *negative* (no such action). Gaze, hesitations, and speech are cues, not labels. Each window ends 16 frames (≈ 1 s at 15 fps) before an interaction onset (pick or place something on the tray) for positives, largest apparent size (closest approach) for negatives : so negatives are harder samples rather than just people walking away.

B. Human baseline

Six annotators judged the same 100 tracks under two conditions: (1) **Pose**, a pose sequence overlaid on a static RGB scene, and (2) **V_{bbox}**, egocentric video with a bounding box on the target person. They were not required to complete both: three labeled both, two only **V_{bbox}**, and one only **Pose** (hence $n=4$ and $n=5$ in Table I). Each completed condition covers all 100 tracks, we do not omit trials that were hard to judge.

For each clip they answered: *Will the person highlighted in the video take or place something on the robot’s tray (camera wearer) within the next 2 seconds?* Clips were shown in randomized order. The static RGB context in **Pose** is required because the image is an equirectangular 360° projection: it grounds the skeleton in the scene (a corridor or hallway).

C. Baseline skeleton models

Table I reports scores for models trained on HUI360-Train with **Pose** input and evaluated on the 100-track subset: (1) LSTM, (2) ST-GCN, and (3) SkateFormer. All models take a

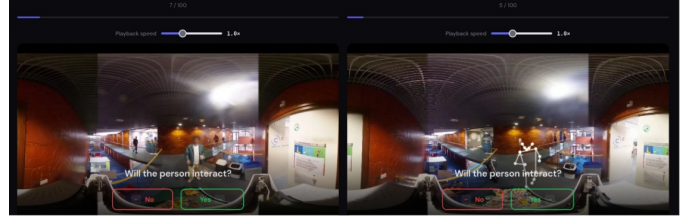


Fig. 1. Human annotation interface for the two evaluation conditions: **V_{bbox}** (left, egocentric video) and **Pose** (right, skeleton overlay on static scene context). Annotators answer whether the highlighted person will interact within the next 2 s.

fixed 32-frame sequence (at 15 fps) of 17 COCO body joints with three channels per joint (2D position and confidence). (1) The LSTM baseline is a three-layer recurrent network (hidden size 128), resulting in a parameter count of just 0.37M. (2) ST-GCN [2] is a spatial-temporal graph convolutional network operating on the OpenPose skeleton graph with learnable edge-importance weighting, we used the default implementation from the official repository and convert our skeletons to the OpenPose format, ST-GCN has 3.07M trainable parameters. (3) SkateFormer [3] is a skeletal-temporal transformer that applies self-attention over explicit spatial partitions of the skeleton with a parameter count of 1.91M.

Training follows the open-source HUI360 protocol with a 16-frame anticipation cutoff (T_{adv}), weighted binary cross-entropy loss, AdamW optimizer (learning rate of 10^{-3} , batch size is 128). LSTM and ST-GCN train for 100 epochs (weight decay 5×10^{-5} and 0.1, respectively); SkateFormer trains for 600 epochs with weight decay 0.1, and uses gradient clipping. ST-GCN and SkateFormer use image-normalized joint coordinates without global standardization; the LSTM additionally normalizes joints within each bounding box and takes globally normalized bounding box coordinates as input.

Table I reports the F1-score, accuracy, precision and recall (at 0.5 probability threshold), as well as per-sequence latency on an RTX 2000 Ada GPU for a full 32-frame sequence, excluding 2D-pose extraction, which we assume feasible in real time at 15 fps with off-the-shelf estimators such as ViTPose [12] or YOLO26-Pose [14].

D. Vision-language models

We evaluate the open-weights Qwen3.5-0.8B, InternVL3-1B, Qwen3.5-35B-A3B, Qwen3.8-Max models as well as the closed ones Gemini 3.5-Flash-Lite, and Gemini 3.6-Flash under three input conditions (denoted I_c , $V + I_c$, and **V_{bbox}**):

- 1) I_c : a cropped image of the target person at the last visible frame (16 frames before onset). Since we want a single image input, we assume the last image before the cutoff is the most meaningful for the model to estimate if someone will interact.
- 2) $V + I_c$: the full equirectangular video and a reference image of the target person; the model is asked to reason about the person shown in the image. Empirically, all models we tested succeeded at tracking the right target

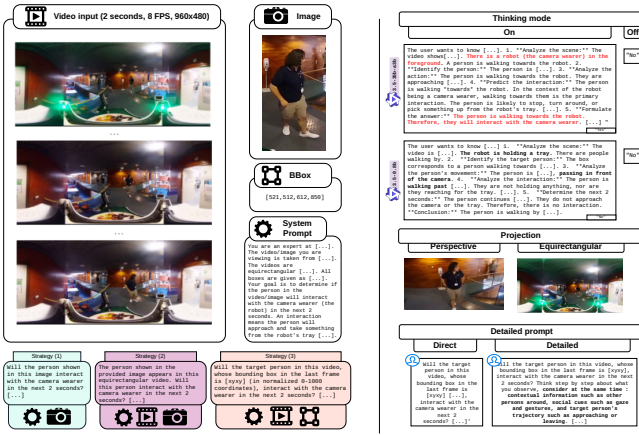


Fig. 2. VLM prompting setup (left) and ablation illustration (right).

person across frames, making this an adequate way of disambiguating multi-person scenarios. Additionally, as motivated for the I_c modality, this close-up image can be used to identify more subtle cues such as gaze and facial expression.

- 3) V_{bbox} : the full equirectangular video together with normalized (0-999) bounding box coordinates of the target person in the last frame. With this modality we exploit the “visual prompting” capacities of recent VLMs to refer to an object by passing normalized coordinates.

Unless noted otherwise, all VLM runs use 8 FPS video sampling from 960x480 equirectangular videos but image crops are made directly on high resolution 3840x1920 equirectangular videos. All models are used with thinking mode enabled and a thinking budget of 1024 tokens. API models are queried via DashScope (Qwen) and Google Gemini, on-device models are served with vLLM on an RTX 2000 Ada GPU.

We report per-track latency: **Pose** models inference time on a 32-frame sequence (excluding 2D-pose extraction); VLMs as mean API or on-device processing time per track.

IV. PRELIMINARY RESULTS

Table I summarizes accuracy, precision, recall, and F1-score on the 100-track subset.

Pose modality. ST-GCN achieves the best F1-score (0.49), ahead of SkateFormer (0.46) and LSTM (0.41). Human annotators with the same modalities reach a mean F1 of 0.57 ± 0.02 (best: 0.59), outperforming all trained pose models showing the ability of humans to infer subtle cues even from the coarse body dynamics alone.

Visual modalities. **Pose** already encodes gait and approach. Video additionally makes nonverbal cues available: gaze toward the tray, head and body orientation, occupied hands, and whether the trajectory is a pass-by or a stop. Humans use these cues, reaching mean F1 0.77 ± 0.08 under V_{bbox} (best: 0.90). VLMs given the same video often do not: the strongest zero-shot result is Qwen3.5-35B-A3B on a single crop I_c (F1 0.57), and several larger models *drop*

when temporal context is added, which is counter-intuitive for anticipation. As illustrated in Fig. 4, video traces frequently reduce the task to whether the person is approaching, yielding high-recall, low-precision predictions : blatant for InternVL3-1B and still visible for Qwen3.8-Max and Gemini 3.6-Flash under V_{bbox} . It is obvious that the video contains more information than the skeleton, and it is confirmed, and quantified, independantly of the models by the human annotators. Then the underperformance of the VLMs highlight a important limitation of current models to reason simultaneously about temporal, spatial and social cues.

Across Table I, smaller on-device models (Qwen3.5-0.8B, InternVL3-1B) score poorly on I_c (F1 0.23–0.35), showing an inability to process visual social cues, but recover more information from dynamics with video (F1 0.40-0.44 under V_{bbox}). Video does not consistently help larger models: Gemini 3.6-Flash peaks on $V + I_c$ (F1 0.53), whereas Qwen3.5-35B-A3B and Qwen3.8-Max perform best with I_c .

TABLE I

METRICS AND HUMAN BASELINES ON 100 HUI360-TEST TRACKS (25 POS. / 75 NEG.): ACCURACY (ACC), PRECISION (PREC), RECALL (REC), AND F1-SCORE. VLM OUTPUT TOKENS INCLUDE THINKING TOKENS; LATENCY IN SECONDS PER TRACK (EXCLUDING 2D-POSE EXTRACTION).

Method	Input	Acc	Prec	Rec	F1	Params	Lat.	Tok.
Humans ($n=4$)	Pose	0.79	0.58	0.57	0.57	—	4.3	—
Humans ($n=5$)	V_{bbox}	0.87	0.71	0.86	0.77	—	4.4	—
LSTM	Pose	0.71	0.42	0.40	0.41	0.37M	0.001	—
ST-GCN	Pose	0.63	0.38	0.72	0.49	3.07M	0.005	—
SkateFormer	Pose	0.42	0.30	1.00	0.46	1.91M	0.017	—
Qwen3.5-0.8B	I_c	0.66	0.26	0.20	0.23	0.8B	16.1	611
Qwen3.5-0.8B	$V + I_c$	0.69	0.25	0.12	0.16	0.8B	12.7	481
Qwen3.5-0.8B	V_{bbox}	0.57	0.33	0.68	0.44	0.8B	8.7	300
InternVL3-1B	I_c	0.41	0.24	0.64	0.35	1B	4.7	279
InternVL3-1B	$V + I_c$	0.29	0.26	1.00	0.41	1B	3.2	82
InternVL3-1B	V_{bbox}	0.25	0.25	1.00	0.40	1B	2.7	80
Qwen3.5-35B-A3B	I_c	0.79	0.58	0.56	0.57	35B	3.4	521
Qwen3.5-35B-A3B	$V + I_c$	0.71	0.44	0.64	0.52	35B	5.2	649
Qwen3.5-35B-A3B	V_{bbox}	0.60	0.35	0.72	0.47	35B	6.2	819
Qwen3.8-Max	I_c	0.70	0.42	0.56	0.48	2.4T	11.7	535
Qwen3.8-Max	$V + I_c$	0.63	0.35	0.56	0.43	2.4T	17.0	699
Qwen3.8-Max	V_{bbox}	0.54	0.33	0.84	0.48	2.4T	16.3	684
Gemini 3.5-Flash-L	I_c	0.73	0.46	0.52	0.49	—	2.5	135
Gemini 3.5-Flash-L	$V + I_c$	0.65	0.37	0.56	0.44	—	3.6	144
Gemini 3.5-Flash-L	V_{bbox}	0.56	0.31	0.60	0.41	—	3.7	172
Gemini 3.6-Flash	I_c	0.69	0.42	0.64	0.51	—	4.7	267
Gemini 3.6-Flash	$V + I_c$	0.68	0.42	0.72	0.53	—	6.5	367
Gemini 3.6-Flash	V_{bbox}	0.53	0.31	0.72	0.43	—	6.1	363

The Pearson correlation (r) between all methods (Fig. 3) reflects the difference between larger and smaller models when analyzing the image and the subtle cues: large models show a relatively high positive r in the range 0.38–0.62 (same family) or 0.32 – 0.78 (across families). Under the V_{bbox} modality, as the focus is given on higher level analysis of the approach motion, the correlation remains positive ($r \in [0.05, 0.21]$) between larger and smaller models (InternVL3-1B and Qwen3.5-0.8B). It drops for Qwen3.5-0.8B and even turns negative for

InternVL3-1B under $\mathbf{I}_c$ and $\mathbf{V} + \mathbf{I}_c$ ($r \in [-0.24, 0.11]$). This trend of motion cues leading to more agreement than social visual cues also holds for humans, whose average r is 0.42 under the skeletal modality and 0.26 under the visual one, highlighting a higher individual bias when human annotators have access to the visual appearance of the target person.

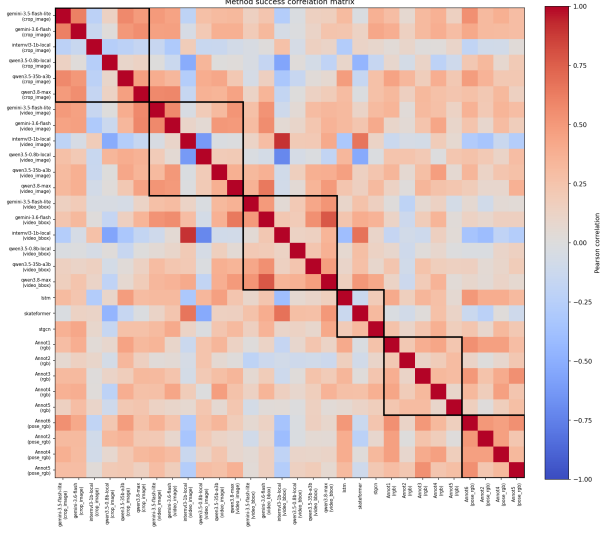


Fig. 3. Pearson correlation (r) between the predictions of all methods and human annotators. Diagonal blocks highlight the correlations between the same family of methods under the same modalities.

Ablations. Table II reports targeted comparisons on Qwen3.5-0.8B and Qwen3.5-35B-A3B under $\mathbf{V}_{\text{bbox}}$. Disabling thinking mode confirms its necessity, as it yields near-zero F1 because models default to almost always answering “No” with very short outputs. Increasing video sampling from 2 to 8 FPS improves F1 monotonically for both models. Equirectangular input consistently outperforms a pinhole-camera style reprojection (90×60 degrees centered on the target person) of the same clip suggesting that the whole visual context, and in particular having the target interaction zone visible is necessary. To encourage the model to reason over both dynamics and social cues we elaborated a longer prompt (detailed) but under thinking mode it has a modest effect, and the short format is slightly preferred.

We saw that the thinking process for Qwen3.8-Max was occasionally interrupted so we vary the reasoning token budget under $\mathbf{V}_{\text{bbox}}$. Increasing the budget from 1024 to 8192 tokens raises F1 from 0.48 to 0.51 and recall from 0.84 to 0.96.

V. CONCLUSION AND DISCUSSION

Humans are able to anticipate interaction from egocentric cues with little effort although not perfectly: they outperform pose models on Pose alone and gain substantially from access to the full video, whereas VLMs struggle to reason over social dynamics and often reduce the task to approach detection or focus on irrelevant details despite thinking mode. We observe no clear correlation between model size and performance, and trained pose models such as ST-GCN remain

TABLE II
VLM ABLATIONS ON $\mathbf{V}_{\text{bbox}}$ (100 TRACKS). SETTINGS MARKED WITH * ARE DEFAULTS. FOR QWEN3.8-MAX (BOTTOM BLOCK), LATENCY AND TOKEN COLUMNS REPORT MEAN $\pm$ STD.

Ablation	Setting	Qwen3.5-0.8B				Qwen3.5-35B-A3B			
		F1	Rec	Lat.	Tok.	F1	Rec	Lat.	Tok.
Thinking	without	0.08	0.04	0.2	3	0.42	0.36	1.4	3
	with*	0.44	0.68	8.7	300	0.47	0.72	6.2	819
FPS	2	0.30	0.40	13.1	345	0.37	0.44	4.7	754
	4	0.30	0.40	9.0	345	0.39	0.52	5.0	784
	8*	0.44	0.68	8.7	300	0.47	0.72	6.2	819
Prompt	detailed	0.36	0.56	14.4	559	0.47	0.76	6.7	960
	direct*	0.44	0.68	8.7	300	0.47	0.72	6.2	819
Projection	pinhole	0.16	0.12	11.2	383	0.38	0.44	5.0	610
	equirectangular*	0.44	0.68	8.7	300	0.47	0.72	6.2	819
Qwen3.8-Max									
Thinking budget		F1	Rec	Lat.	Tok.				
	1024*	0.48	0.84	16.3 $\pm$ 6.4	684 $\pm$ 315				
	2048	0.49	0.88	23.3 $\pm$ 12.0	908 $\pm$ 636				
	8192	0.51	0.96	24.5 $\pm$ 22.3	1101 $\pm$ 1240				



Fig. 4. Qualitative examples of successful and unsuccessful predictions. Highlighted elements in the VLMs answers are motion-related thinking (**bold, italic**), visual-social related (**bold, underline**), and other noticeable recurring elements in color: certainty/uncertainty assessment (light blue), explicitly refer to related datasets (dark blue), reason about hands/arms positions (pink).

an acceptable lightweight alternative when compute is limited. These results are preliminary; future work should compare VLMs to lighter image- or video-based models trained on HUI360, probe whether failures occur at social perception or at the anticipation decision, and conduct a deeper user study to separate rational inference from intuitive judgment in human annotators.

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REFERENCES

- [1] R. Lorenzo-Louis, F. Amadio, B. Luvison, and S. Ivaldi, "HUI360: A 360° egocentric dataset and baselines for human-robot interaction anticipation," in *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, Kyoto, Japan, 2026, hAL hal-05609928. [Online]. Available: <https://hucebot.github.io/hui360>
- [2] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.
- [3] J. Do and M. Kim, "Skateformer: skeletal-temporal transformer for human action recognition," in *European Conference on Computer Vision*. Springer, 2025, pp. 401–420.
- [4] S. Thompson, A. Lew, Y. Li, E. Stanish, A. Huang, R. Phanse, and M. Vázquez, "Predicting human intent to interact with a public robot: The people approaching robots database (PAR-D)," in *Proceedings of the 26th International Conference on Multimodal Interaction (ICMI)*. San Jose, Costa Rica: ACM, 2024, pp. 536–545. [Online]. Available: <https://doi.org/10.1145/3678957.3685706>
- [5] Y. Kato, T. Kanda, and H. Ishiguro, "May i help you? - design of human-like polite approaching behavior-," in *2015 10th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2015, pp. 35–42.
- [6] F. Bu, M. Tsai, A. Tjokro, T. Bhattacharjee, J. Ortiz, and W. Ju, "Using vision-language models as proxies for social intelligence in human-robot interaction," 2025. [Online]. Available: <https://arxiv.org/abs/2512.07177>
- [7] D. Song, J. Liang, A. Payandeh, A. H. Raj, X. Xiao, and D. Manocha, "VLM-Social-Nav: Socially aware robot navigation through scoring using vision-language models," 2024. [Online]. Available: <https://arxiv.org/abs/2404.00210>
- [8] S. Bhattacharyya and J. Z. Wang, "Evaluating vision-language models for emotion recognition," 2025. [Online]. Available: <https://arxiv.org/abs/2502.05660>
- [9] M. Agarwal, S. A. Tsafaris, L. Sevilla-Lara, and S. McDonagh, "Why do vision language models struggle to recognize human emotions?" 2026. [Online]. Available: <https://arxiv.org/abs/2604.15280>
- [10] M. T. Parreira, H. Quan, A. G. Ramirez-Aristizabal, and W. Ju, "Bad idea or good prediction? comparing VLM and human anticipatory judgment," in *Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI Companion)*. Edinburgh, Scotland, UK: ACM, 2026, pp. 933–937.
- [11] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rdlc, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollr, and C. Feichtenhofer, "Sam 2: Segment anything in images and videos," 2024. [Online]. Available: <https://arxiv.org/abs/2408.00714>
- [12] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "ViTPose: Simple vision transformer baselines for human pose estimation," in *Advances in Neural Information Processing Systems*, 2022.
- [13] R. Khirodkar, T. Bagautdinov, J. Martinez, S. Zhaoen, A. James, P. Selednik, S. Anderson, and S. Saito, "Sapiens: Foundation for human vision models," *arXiv preprint arXiv:2408.12569*, 2024.
- [14] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics yolo26: Unified real-time end-to-end vision models," 2026. [Online]. Available: <https://arxiv.org/abs/2606.03748>